

# Notes

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## Logarithm

### Domain

$(0, \infty)$

### Range

$(-\infty, \infty)$

## Polynomials

### Factor theorem

The factor theorem states that  $(x - r)$  is a factor of a polynomial  $p(x)$  if and only if the root  $x = r$  is also a root of  $p(x)$ .

### Exercises

- The polynomial  $-5x^4 + 8x^3 + 29x^2 - 20x - 12$  can be expressed as  $-(x - 1)(x - 3) \cdot P(x)$ . Find  $P(x)$ .

First let's divide by  $(x - 1)$

$$\begin{array}{r|rrrrrr} 1 & -5 & 8 & 29 & -20 & -12 \\ & & -5 & 3 & 32 & 12 \\ \hline & -5 & 3 & 32 & 12 & 0 \end{array}$$

So we have  $(x - 1) \cdot (-5x^3 + 3x^2 + 32x + 12)$

Let's divide by  $(x - 3)$

$$\begin{array}{r|rrrr} 3 & -5 & 3 & 32 & 12 \\ & & -15 & -36 & -12 \\ \hline & -5 & -12 & -4 & 0 \end{array}$$

So we have  $(x - 1)(x - 3)(-5x^2 - 12x - 4)$

So we have  $(x - 1) \cdot (x - 3) \cdot (-5x^2 - 2x + 4)$

Finally,  $-(x - 1)(x - 3)(5x^2 + 2x + 4)$ . So  $P(x) = 5x^2 + 2x + 4$ .

Factoring further,  $P(x) = (5x + 2)(x + 2)$ .

- Find the polynomial  $P(x)$  given that

$$4x^4 - 19x^3 - 9x^2 + 19x + 5 = (x - 1)(x - 5)P(x)$$

First let's divide by  $(x - 1)$ .

$$\begin{array}{r|rrrrr} 1 & 4 & -19 & -9 & 19 & 5 \\ & & 4 & -15 & -24 & -5 \\ \hline & 4 & -15 & -24 & -5 & 0 \end{array}$$

So  $(x - 1)(4x^3 - 15x^2 - 24x - 5)$ . Now divide by  $(x - 5)$ .

$$\begin{array}{r|rrrr} 5 & 4 & -15 & -24 & -5 \\ & & 20 & 25 & 5 \\ \hline & 4 & 5 & 1 & 0 \end{array}$$

So  $(x - 1)(x - 5)(4x^2 + 5x + 1)$ .

## Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} \\ &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \frac{1 - \sqrt{2}}{-1} \\ &= -1 + \sqrt{2} \end{aligned}$$

## Exercises

•

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} \\ &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \frac{1 - \sqrt{2}}{-1} \\ &= -1 + \sqrt{2} \end{aligned}$$

1.

$$\begin{aligned}
& \frac{1 + \sqrt{8}}{\sqrt{9} - \sqrt{2}} \\
&= \frac{(1 + \sqrt{8})(\sqrt{9} + \sqrt{2})}{7} \\
&= \frac{\sqrt{9} + 3\sqrt{8} + \sqrt{2} + \sqrt{16}}{7} \\
&= \frac{3 + 6\sqrt{2} + \sqrt{2} + 4}{7} \\
&= \frac{7 + 7\sqrt{2}}{7} \\
&= 1 + \sqrt{2}
\end{aligned}$$

2.

$$\begin{aligned}
\frac{x^2 y^2}{\sqrt{x^3 y^3}} &= \frac{x^2 y^2 \sqrt{x^3 y^3}}{x^3 y^3} \\
&= \frac{x^2 y^2 x^{3/2} y^{3/2}}{x^3 y^3} \\
&= \frac{x^{7/2} y^{7/2}}{x^3 y^3} \\
&= \sqrt{xy}
\end{aligned}$$

3.

$$\begin{aligned}
\frac{\sqrt{12x}}{3\sqrt{2x^2y}} &= \frac{\sqrt{12x} \cdot \sqrt{2x^2y}}{3 \cdot 2x^2y} \\
&= \frac{\sqrt{24x^3y}}{6x^2y} \\
&= \frac{\sqrt{4x^2 \cdot 6xy}}{6x^2y} \\
&= \frac{2x \cdot \sqrt{6xy}}{6x^2y} \\
&= \frac{\sqrt{6xy}}{3xy}
\end{aligned}$$

4. Given that  $d > 0$ .

$$\begin{aligned}
 \frac{cd}{\sqrt{cd^3}} &= \frac{cd\sqrt{cd^3}}{cd^3} \\
 &= \frac{\sqrt{cd^3}}{d^2} \\
 &= \frac{\sqrt{c} \cdot d^{3/2}}{d^2} \\
 &= \frac{\sqrt{c} \cdot d^{1/2}}{d} \\
 &= \frac{\sqrt{cd}}{d}
 \end{aligned}$$

5. Given that  $p > 0$

$$\begin{aligned}
 \frac{7t}{\sqrt{14tp^2}} &= \frac{7t \cdot \sqrt{14tp^2}}{14tp^2} \\
 &= \frac{7tp\sqrt{14t}}{14tp^2} \\
 &= \frac{\sqrt{14t}}{2p}
 \end{aligned}$$

6. Given that  $a > 0$

$$\begin{aligned}
 \sqrt{\frac{5b}{a^3}} &= \frac{\sqrt{5b}}{\sqrt{a^3}} \\
 &= \frac{\sqrt{5b}}{a\sqrt{a}} \\
 &= \frac{\sqrt{5ab}}{a\sqrt{a}^2} \\
 &= \frac{\sqrt{5ab}}{a^2}
 \end{aligned}$$

7.

$$\begin{aligned}
 \sqrt{\frac{14r}{p^5}} &= \frac{\sqrt{14r}}{\sqrt{p^5}} \\
 &= \frac{\sqrt{14p^5r}}{p^5} \\
 &= \frac{\sqrt{14r} \cdot p^{5/2}}{p^5} \\
 &= \frac{\sqrt{14pr}}{p^3}
 \end{aligned}$$

8.

$$\begin{aligned}
 \sqrt{\frac{9a^2}{b^3}} &= \frac{\sqrt{9a^2}}{\sqrt{b^3}} \\
 &= \frac{\sqrt{9a^2}}{b\sqrt{b}} \\
 &= \frac{\sqrt{9a^2b}}{b\sqrt{b}^2} \\
 &= \frac{3a\sqrt{b}}{b^2}
 \end{aligned}$$

9. Given that  $z > 0$ .

$$\begin{aligned}
 \sqrt{\frac{11}{wz^2}} &= \frac{\sqrt{11}}{\sqrt{wz^2}} \mathfrak{s} = \frac{\sqrt{11}}{z\sqrt{w}} \\
 &= \frac{\sqrt{11w}}{zw}
 \end{aligned}$$

10.

$$\begin{aligned}
 \sqrt{\frac{5a}{3b}} &= \frac{\sqrt{5a}}{\sqrt{3b}} \\
 &= \frac{\sqrt{15ab}}{3b}
 \end{aligned}$$

11.

$$\begin{aligned}
 \sqrt{\frac{a^2+3}{a^2-3}} &= \frac{\sqrt{a^2+3}}{\sqrt{a^2-3}} \\
 &= \frac{\sqrt{(a^2+3)(a^2-3)}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3}
 \end{aligned}$$

12. Given  $y > 0$

$$\begin{aligned}
 \frac{2+3\sqrt{xy}}{3\sqrt{y}} &= \frac{(2+3\sqrt{xy}) \cdot \sqrt{y}}{3y} \\
 &= \frac{3y\sqrt{x}+2\sqrt{y}}{3y} \\
 &= \sqrt{x} + \frac{2\sqrt{y}}{3y}
 \end{aligned}$$

## Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x-1)(3x-2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x-1)(3x-2)} = \frac{A}{x-1} + \frac{B}{3x-2}$$

Multiplying both sides by  $(x-1)(3x-2)$ , we get:

$$1 = A(3x-2) + B(x-1)$$

Setting  $x = 1$ , we get:

$$1 = A(3(1)-2) + B(1-1) = A$$

Setting  $x = \frac{2}{3}$ , we get:

$$1 = A\left(3\left(\frac{2}{3}\right) - 2\right) + B\left(\frac{2}{3} - 1\right) = -\frac{B}{3}$$

Solving for  $B$ , we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

### Exercises

1. Decompose  $\frac{44x+48}{(x+6)(x-3)(x+2)}$

$$\frac{44x+48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$

$$44x+48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set  $x = 3$ :

$$44 \cdot 3 + 48 = B \cdot 9 \cdot 5$$

$$\frac{180}{9 \cdot 5} = B$$

$$B = 4$$

Set  $x = -2$ :

$$44 \cdot -2 + 48 = C \cdot 4 \cdot -5$$

$$-40 = C \cdot -20$$

$$C = 2$$

Set  $x = -6$ :

$$44 \cdot -6 + 48 = A \cdot -9 \cdot -4$$

$$A = -6$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find  $B$

$$\begin{aligned}\frac{-2x-9}{x+3}x &= \frac{1}{x+3} + \frac{B}{x} \\ (-2x-9) &= x + B(x+3) \\ -3x-9 &= B(x+3) \\ \frac{-3(x+3)}{x+3} &= B \\ B &= -3\end{aligned}$$

3. Given that

$$\begin{aligned}\frac{-9x+36}{(x-3)(x-6)} &= -\frac{3}{x-3} + \frac{B}{x-6} \\ -9x+36 &= -3(x-6) + B(x-3) \\ -6x+18 &= B(x-3) \\ \frac{-6(x-3)}{x-3} &= B \\ B &= -6\end{aligned}$$

## Functions

### Horizontal asymptotes of rational functions

Rational functions are of the form:

$$f(x) = \frac{ax^n + bx^{n-1} + \dots}{cx^m + dx^{m-1} + \dots}$$

To get the horizontal asymptotes, you first take the **dominant term**. It's the leading term of the polynomial in the denominator, *without the coefficient*. Here  $x^m$ .

Then, you divide every term in both numerator and denominator, and evaluate at  $x \rightarrow \infty$ .

### Radical functions

#### Domain

All the values where the radicand  $> 0$  for even degrees,  $(-\infty, \infty)$  for odd degrees.

#### Exercises

- Domain of  $d(x) = \sqrt[7]{-2x + \frac{1}{2}} + 1$ ?

$$-2x + \frac{1}{2} >$$

### Roots of rational functions

First factor numerator and denominator, then cancel out common factors, then set the numerator to zero and solve.

#### Exercises

- Solve  $f(t) = 0$  where  $f(t) = \frac{7t^2 + 14}{t^2 - 25}$ .



$$\frac{7t^2 + 14}{t^2 - 25} = 0$$

$$\frac{7(t^2 + 2)}{(t + 5)(t - 5)} = 0$$

$$t^2 + 2 = 0$$

$$t^2 = -2$$

2. Solve  $f(x) = 0$  where  $f(x) = \frac{x^2+2x-15}{x^3+5x^2-9x-45}$ .

$$\frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45} = 0$$

$$\frac{x^2 + 5x - 3x - 15}{x^2(x + 5) - 9(x + 5)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x^2 - 9)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x - 3)(x + 3)} = 0$$

$$1 = 0$$

The equation has no solution.

3. Solve  $f(t) = 0$  where  $f(t) = \frac{t-3}{t^2+6t+5}$

$$\frac{t - 3}{t^2 + 6t + 5} = 0$$

$$\frac{t - 3}{t^2 + t + 5t + 5} = 0$$

$$\frac{t - 3}{(t + 5)(t + 1)} = 0$$

$$t - 3 = 0$$

$$t = 3$$

## Invertible functions

A function is invertible on an interval if and only if it is one-to-one on that interval.

To check if a function is one-to-one, we can use the horizontal line test. If any horizontal line intersects the graph of the function at most once, then the function is one-to-one.

## Statistics

### Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

### Covariance

Covariance is the measure of the joint variability of two random variables.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

## Mean estimation for continuous grouped data

$$\bar{x} \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot m_i$$

where

- $f_i$  is the frequency of the  $i$ th interval
- $m_i$  is the midpoint of the  $i$ th interval
- $n$  is the total number of observations

## Estimating a variance for continuous grouped data

$$\sigma_n^2 \approx \frac{1}{n} \sum_{i=1}^K (m_i - \bar{x})^2$$

## Graphing

### Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

#### Example

Given that  $(x - 3)$  is a factor of  $f(x) = x^3 + x^2 - 4x - 24$ , sketch the curve of  $y = f(x)$ .

First, synthetic division:

|   | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|---|-------|-------|-------|-------|
| 3 | 1     | 1     | -4    | -24   |
|   |       | 3     | 12    | 24    |
|   | 1     | 4     | 8     | 0     |

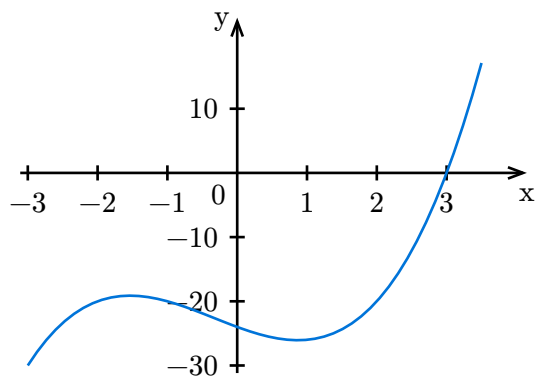
So we have  $f(x) = (x - 3)(x^2 + 4x + 8)$

$\therefore x = 3$  is a root

The discriminant of  $x^2 + 4x + 8$  is  $4^2 - 4 \cdot 1 \cdot 8 = -16$ , which is negative, so  $x = 3$  is the only real root.

The  $y$  intercept is  $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



## Logarithms

### Domain

$(0, \infty)$

### Range

$(-\infty, \infty)$

## Complex Numbers

### Magnitude

The magnitude of a complex number  $z = a + bi$  is defined as  $|z| = \sqrt{a^2 + b^2}$ .

### Argument

For any complex number, like  $z = 3 + 4i$ , there is  $\arg(z)$  that is the angle (in radians) between the  $x$  axis and the complex number in the Argand diagram.

So for  $z$ :

$$\begin{aligned}\tan \theta &= \frac{4}{3} \\ \theta &= \arctan\left(\frac{4}{3}\right) \\ &\approx 0.93\end{aligned}$$

## Trigonometry

### Unit circle

#### Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the  $x$ -axis. It is always a positive angle less than or equal to 90 degrees.

#### Pythagorean identity in the first quadrant

In the first quadrant, any point  $(x, y)$  forms a right triangle with the origin, where  $x$  and  $y$  are the lengths of the legs and  $\sqrt{x^2 + y^2}$  is the length of the hypotenuse. Since in the unit circle, the length of the hypotenuse is 1, by definition, the Pythagorean identity states that  $x^2 + y^2 = 1$ .

### Exercises

1. For a point in the first quadrant with  $y = \frac{\sqrt{5}}{3}$ , find its  $x$  coordinate.

$$\sqrt{\left(\frac{\sqrt{5}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{5}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{5}}{3}\right)^2$$

$$x^2 = 1 - \frac{5}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \sqrt{\frac{4}{9}}$$

$$x = \frac{2}{3}$$

2. Point in the first quadrant.  $y$  coordinate is  $\frac{\sqrt{2}}{3}$ . What is the  $x$  coordinate?

$$\sqrt{\left(\frac{\sqrt{2}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{2}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{2}}{3}\right)^2$$

$$x^2 = 1 - \frac{2}{9}$$

$$x^2 = \frac{7}{9}$$

$$x = \sqrt{\frac{7}{9}}$$

$$x = \frac{\sqrt{7}}{3}$$

### Pythagorean trigonometric identity

In the unit circle, since for any central angle  $x = \cos \theta$  and  $y = \sin \theta$ .

That means

$$x^2 + y^2 = 1$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This is the **Pythagorean trigonometric identity**. It works in all quadrants.

### Exercises

1. Point  $P$  on the unit circle with  $x = \frac{2}{5}$ . Find  $\sin \theta$ .

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \frac{4}{25}}$$

$$\sin \theta = \sqrt{\frac{21}{25}}$$

$$\sin \theta = \frac{\sqrt{21}}{5}$$

2.  $y$  coordinate is  $\frac{4}{9}$ . Find  $\cos \theta$ .

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{4}{9}\right)^2 + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \left(\frac{4}{9}\right)^2$$

$$\cos^2 \theta = 1 - \frac{16}{81}$$

$$\cos^2 \theta = \frac{65}{81}$$

$$\cos \theta = \sqrt{\frac{65}{81}}$$

$$\cos \theta = \frac{\sqrt{65}}{9}$$

3.  $x$  coordinate is  $\frac{\sqrt{7}}{4}$ . Find  $\csc \theta$ .

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\sin \theta = \sqrt{1 - \frac{7}{16}}$$

$$\sin \theta = \sqrt{\frac{9}{16}}$$

$$\sin \theta = \frac{3}{4}$$

$\therefore$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{4}{3}$$

4.  $x$  coordinate is  $\frac{3}{7}$ . Find  $\csc \theta$ .

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \frac{9}{49}}$$

$$\sin \theta = \sqrt{\frac{40}{49}}$$

$$\sin \theta = \frac{\sqrt{40}}{7}$$

$\therefore$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{7}{\sqrt{40}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7\sqrt{10}}{20}$$

5.  $y$  coordinate is  $\frac{1}{9}$ . Find  $\tan \theta$ .

We have  $\sin \theta = \frac{1}{9}$ .

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{1}{9}\right)^2}$$

$$\cos \theta = \sqrt{1 - \frac{1}{81}}$$

$$\cos \theta = \sqrt{\frac{80}{81}}$$

$$\cos \theta = \frac{\sqrt{80}}{9}$$

$$\cos \theta = \frac{4\sqrt{5}}{9}$$

$\therefore$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{\frac{1}{9}}{\frac{4\sqrt{5}}{9}}$$

$$\tan \theta = \frac{1}{9} \cdot \frac{9}{4\sqrt{5}}$$

$$\tan \theta = \frac{\sqrt{5}}{20}$$

6.  $x$  coordinate is  $\frac{\sqrt{11}}{6}$ . Find  $\cot(\theta)$

We have  $\cos \theta = \frac{\sqrt{11}}{6}$  and  $\cot(\theta) = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\frac{\sqrt{11}}{6}}{\sin \theta}$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin^2 \theta = 1 - \left(\frac{\sqrt{11}}{6}\right)^2$$

$$\sin^2 \theta = 1 - \frac{11}{36}$$

$$\sin^2 \theta = \frac{25}{36}$$

$$\sin \theta = \sqrt{\frac{25}{36}}$$

$$\sin \theta = \frac{5}{6}$$

$\therefore$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{\frac{\sqrt{11}}{6}}{\frac{5}{6}}$$

$$\cot \theta = \frac{\sqrt{11}}{5}$$

### Quadrantal angles

Quadrantal angles are angles that are multiples of  $90^\circ$ . For example,  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ ,  $270^\circ$ , and  $360^\circ$  are all quadrantal angles. They lie on an axis.

They are useful to find the values of trigonometric functions. Just remember that for points on the unit circle,  $\cos \theta = x$  and  $\sin \theta = y$ .

The corresponding angles on  $[0, 360^\circ]$  are  $0, \frac{\pi}{2}, \pi, 3\frac{\pi}{2}, 2\pi$ .

Example:  $\cos 0 + \sin\left(\frac{\pi}{2}\right) = 1 + 1 = 2$

### Examples

- Find the value of  $\tan(180^\circ) + \cot(270^\circ)$

$$\begin{aligned}\tan(180^\circ) + \cot(270^\circ) &= \frac{\sin(180^\circ)}{\cos(180^\circ)} + \frac{\cos(270^\circ)}{\sin(270^\circ)} \\ &= \frac{0}{-1} + \frac{0}{-1} \\ &= 0\end{aligned}$$

- $\sec\left(\frac{\pi}{2}\right)$ ?

$$\begin{aligned}\sec\left(\frac{\pi}{2}\right) &= \frac{1}{\cos\left(\frac{\pi}{2}\right)} \\ &= \frac{1}{0}\end{aligned}$$

So it's undefined.

## Calculus

### Limits

#### Limits of logarithmic functions

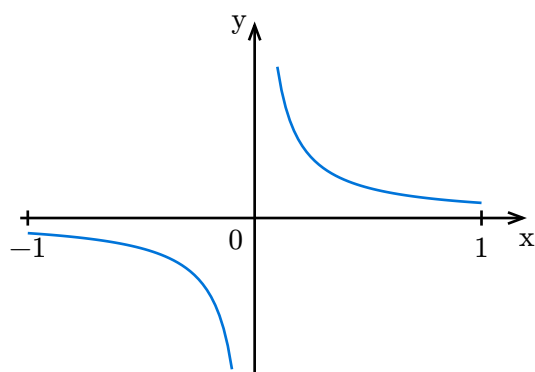
For any real  $n$ :

$$\lim_{x \rightarrow n} \log(x) = \log(x)$$

For infinity, remember the domain and apply transformations.

#### Limits of reciprocal functions

Remember, reciprocal functions have the form  $f(x) = \frac{1}{x}$ . The limit of a reciprocal function as  $x$  approaches infinity is always zero.



In general, for any non-constant polynomial  $P(x)$ , we have  $\lim_{x \rightarrow \pm\infty} \frac{1}{P(x)} = 0$

But for  $\lim(x \rightarrow 0)$ , we have to consider the degree of the leading term of  $P(x)$ .

- Any function of the form  $y = \frac{1}{x^n}$  where  $n$  is odd has the same shape as  $y = \frac{1}{x}$ . There is no limit at  $x = 0$  because left and right limit are not the same.
- Any function of the form  $y = \frac{1}{x^n}$  where  $n$  is even has the same shape as  $y = \frac{1}{x^2}$ . There is a limit, since the function will approach  $\infty$  or  $-\infty$ .

### Differentiation

#### Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x))$$

#### Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for  $y = x^2 + x$ :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 + x) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x) \\ &= 2x + 1 \end{aligned}$$



### The power rule

Given a monomial  $f(x) = x^n$ , where  $n$  is a real number,

$$f'(x) = nx^{n-1}$$

### Exercises

1. Calculate  $\frac{dy}{dx}$  for  $y = 2x - \sqrt{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(2x - \sqrt{x}) \\ &= \frac{d}{dx}2x - \frac{d}{dx}\sqrt{x} \\ &= 2 - \frac{d}{dx}x^{1/2} \\ &= 2 - \frac{1}{2} \cdot x^{-1/2} \\ &= 2 - \frac{1}{2\sqrt{x}}\end{aligned}$$

2. Find the derivative of  $f(x) = \frac{1}{8}x^8$

$$\begin{aligned}f'(x) &= \frac{1}{8} \cdot \frac{d}{dx}x^8 \\ &= \frac{1}{8} \cdot 8x^7 \\ &= \frac{8x^7}{8} \\ &= x^7\end{aligned}$$

3. Derivative of  $f(x) = 8\sqrt{x}$

$$\begin{aligned}f'(x) &= \frac{d}{dx}8\sqrt{x} \\ &= \frac{d}{dx}8x^{1/2} \\ &= 8 \cdot \frac{d}{dx}\left(\frac{1}{2} \cdot x^{-1/2}\right) \\ &= 8 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{4}{\sqrt{x}}\end{aligned}$$

### Rate of change

Average rate of change on an interval  $[a, a + h]$ :

$$\frac{\delta y}{\delta x} = \frac{f(a + h) - f(a)}{h}$$

The *instantaneous rate of change* at a point  $x = a$  is given by:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

## Exercises

1. Given that  $f(x) = x^2$ , what is  $f'(2)$  according to the definition?

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\ &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\ &= \lim_{h \rightarrow 0} (4 + h) \\ &= 4 \end{aligned}$$

## Integration

### The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of  $x^2$  is  $2x$ , we have  $x^2$  as an antiderivative of  $2x$ . But  $x^2 + 2$ ,  $x^2 + 1$ , etc. are also antiderivatives of  $2x$ .

In general, the antiderivatives of  $2x$  follow the pattern  $x^2 + C$  where  $C$  is a constant.

We can also formulate that using the  $\int$  symbol for integration:

$$\int 2x \, dx = x^2 + C$$

Where  $C$  is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here  $2x$ ) is called the **integrand**.

### The power rule for integration

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

## Exercises

1. Antiderivative of  $\frac{1}{x^4}$ ?

$$\begin{aligned} \int \frac{1}{x^4} dx &= \int x^{-4} dx \\ &= \frac{x^{-4+1}}{-4+1} + C \\ &= \frac{x^{-3}}{-3} + C \\ &= -\frac{1}{3}x^3 + C \end{aligned}$$

2. Antiderivative of  $x^{\frac{5}{8}}$

$$\begin{aligned}
 \int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\
 &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\
 &= \frac{8x^{\frac{13}{8}}}{13} + C
 \end{aligned}$$

3. Antiderivative of  $\sqrt[5]{x^7}$

$$\begin{aligned}
 \int x^{\frac{7}{5}} dx &= \frac{x^{7/5+1}}{7/5+1} + C \\
 &= \frac{x^{12/5}}{12/5} \\
 &= \frac{5x^{12/5}}{12}
 \end{aligned}$$

4. Calculate  $\int \left(\frac{z}{5} dz\right)$

$$\begin{aligned}
 \int \left(\frac{z}{5} dz\right) &= \frac{1}{5} \int z dz \\
 &= \frac{1}{5} \cdot \left(\frac{z^2}{2} + C\right) \\
 &= \frac{z^2}{10} + C
 \end{aligned}$$

### The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like  $5x^2$ , we can take the constant factor out of the integral:

$$\int 5x^2 dx = 5 \cdot \int x^2 dx$$

This is the **constant factor rule**. In general:

$$\int k f(x) dx = k \int f(x) dx$$

Where  $k$  is a **constant**. It does not work if  $k$  is a variable.

### Exercises

1. Calculate  $\int \frac{1}{2y^5} dy$

$$\begin{aligned}
 \int \frac{1}{2y^5} dy &= \frac{1}{2} \int y^{-5} \\
 &= \frac{1}{2} \cdot \frac{y^{-5+1}}{-5+1} + C \\
 &= \frac{1}{2} \cdot \frac{y^{-4}}{-4} + C \\
 &= \frac{1}{2} \cdot -\frac{1}{4y^4} + C \\
 &= -\frac{1}{8y^4} + C
 \end{aligned}$$

2. Calculate  $\int \frac{1}{3x^6} dx$

$$\begin{aligned}
 \int \frac{1}{3x^6} dx &= \frac{1}{3} \cdot \int x^{-6} \\
 &= \frac{1}{3} \cdot \frac{x^{-5}}{-5} + C \\
 &= \frac{1}{3} \cdot -\frac{1}{5x^5} + C \\
 &= -\frac{1}{15x^5} + C
 \end{aligned}$$

3. Calculate  $\int \frac{\pi}{2} dz$

$$\begin{aligned}
 \int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\
 &= \frac{\pi}{2} \cdot z + C \\
 &= \frac{\pi z}{2} + C
 \end{aligned}$$

### The sum rule for indefinite integrals

To take the integral of a sum, you can take the sum of the integrals of each term:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

The sum rule can be combined with the constant factor rule to integrate sums of constant multiples of functions:

$$\int (k_1 f(x) + k_2 g(x)) dx = k_1 \int f(x) dx + k_2 \int g(x) dx$$

### Exercises

- Calculate  $\int \sqrt{x}(2+x) dx$

$$\begin{aligned}
\int \sqrt{x}(2+x) \, dx &= \int 2\sqrt{x} + x \cdot \sqrt{x} \, dx \\
&= \int 2\sqrt{x} \, dx + \int x \cdot \sqrt{x} \, dx \\
&= 2 \int \sqrt{x} \, dx + \int x^{3/2} \, dx \\
&= 2 \cdot \frac{x^{3/2}}{\frac{3}{2}} + \frac{x^{5/2}}{\frac{5}{2}} + C \\
&= \frac{4x^{3/2}}{3} + \frac{2x^{5/2}}{5} + C \\
&= \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + C
\end{aligned}$$

- Calculate  $\int \frac{2+x^4}{x^2} \, dx$

$$\begin{aligned}
\int \frac{2+x^4}{x^2} \, dx &= \int \frac{2}{x^2} \, dx + \int \frac{x^4}{x^2} \, dx \\
&= \int 2 \cdot x^{-2} \, dx + \int x^2 \, dx \\
&= 2 \cdot \int x^{-2} \, dx + \frac{x^3}{3} + C \\
&= 2 \cdot -x^{-1} + \frac{x^3}{3} + C
\end{aligned}$$

- Calculate:

$$\begin{aligned}
\int (3-x^2)^2 \, dx &= \int (9-6x^2+x^4) \, dx \\
&= 9 \int 1 \, dx - 6 \int x^2 \, dx + \int x^4 \, dx \\
&= 9x - 6 \frac{x^3}{3} + \frac{x^5}{5} + C \\
&= \frac{x^5}{5} - 2x^3 + 9x + C
\end{aligned}$$

- Find the antiderivative of  $z^3 + 2z - 1$

$$\begin{aligned}
\int z^3 + 2z - 1 \, dz &= \int z^3 \, dz + \int 2z \, dz - \int 1 \, dz \\
&= \frac{z^4}{4} + 2 \frac{z^2}{2} - z + C \\
&= \frac{z^4}{4} + z^2 - z + C
\end{aligned}$$